CS 556: Task-Sheet for Lab 5

Wall Following (PD-Controller)

Name: Colin Treziok & Cameron Lee

Robot ID: 20

Question:	1a Prelab	1b Prelab	1c Prelab	3.1	3.2a	3.2b	3.2c	3.2d	3.2e	3.2f	3.3a	3.3b	3.3c	Total
Points:	5	5	10	20	15	5	5	5	5	5	5	10	5	100
Score:														

Pre-lab Submission (on Canvas, Due Wednesday, 23:59 PM before lab #5):

- 1) Task-Sheet Pre-lab task (one PDF file: TeamX_LabY_Prelab_Tasksheet.pdf)
- 2) Draft code (one zip folder: TeamX_LabY_DraftCodes.zip)

Post-lab submission (On Canvas, Due Wednesday, 23:59 PM after lab #5) Submit in 3 separate files (do NOT zip these four together):

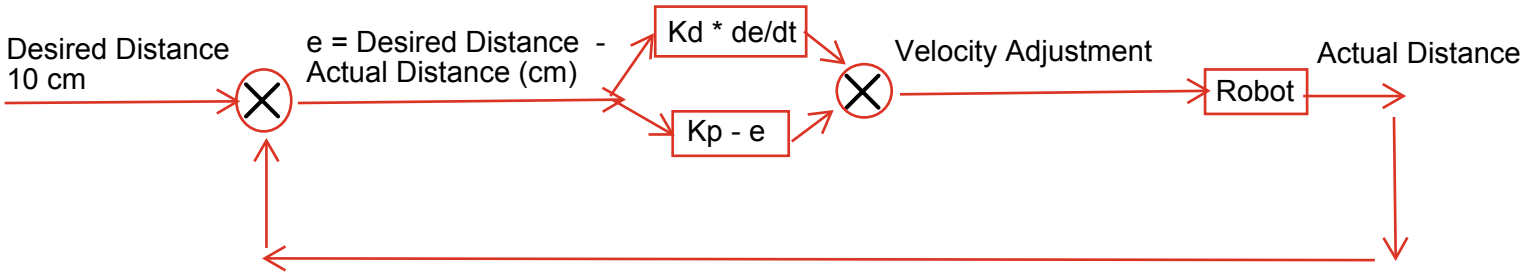
- 1) Task-Sheet tasks (one PDF file: TeamX_LabY_Postlab_Tasksheet.pdf).
- 2) Final code (one ZIP folder: TeamX_LabY_FinalCodes.zip).
- 3) Videos (one zip folder: TeamX_LabY_Videos.zip).

Lab Grading Criteria:

- Every member of the team must submit the same set of final materials for the post-lab to receive a grade.
- For the pre-lab submissions, no extensions will be offered. Failure to prepare for the lab will result in a 20% deduction from your total grade for that week's lab.
- For the post-lab submissions, late work or post-lab submissions will incur a 20% penalty per day, up to a maximum deduction of 100%.
- Lab attendance is mandatory – a missed lab will mean 0 points for you.
- Before you leave, make sure you return all robots and tools used to the front desk in the same format that you received them. If not done so, 10 points will be deducted.
- Your code must follow the industry guidelines for coding; if not done so, 10 points will be deducted. Comment out code for easy review, add new code below, and label each task with clear, descriptive comments. Ensure a modular, well-structured design with meaningful names, minimal globals, and avoid hardcoding values.

1. Prelab

a. (5 points) Draw the block diagram for your proportional derivative PDController. Make sure to label and give units for the goal state, error, and controller output.



b. (5 points) (Controller Architecture)

i. Current state and how to estimate it:

The current state is the actual distance to the wall and is estimated using the sonar sensor.

ii. Goal state:

The goal state is the desired distance to the wall which for this lab is 10 cm from the wall

iii. Function for the right wheel:

The function for the right wheel would be Vright = Vbase - PDout where PDout is the output from the controller or the velocity adujustment

iv. Function for the left wheel:

The function for the left wheel would be Vleft = Vbase + Pdout

c. (10 points) PDController Draft Code

3.1 (20 points) Final PDController class submission - [25 points for working code and 5 points for comments.]

3.2 PD Controller Wall Following

a. (15 points) Wall Following Working [Video for each test (b-d), **3 videos in total**]
b. (5 points) List the first set of gain K_P and K_D, and clamping ranges, and describe the robot's behavior.

First test had Kp 5 and Kd 1 with clamping ranges of -10 and 10, the robot was pretty smooth with little oscillations on the straight and mostly smooth turns, though it did jitter a bit more when turning. On our second test we upped the clamping to -20 and 20 and it was still pretty smooth but overall we noticed it jittering a bit more.

c. (5 points) List the second set of gain K_P and K_D, and clamping ranges, and describe the robot's behavior.

For our next test we changed Kp to 1 and Kd to 5 with clamping ranges of -10 and 10. With this test we saw that the robot was smoother on the straight, but struggled much more on the turns sometimes overshooting it and other times undershooting it. We then changed the clamping to -20 and 20 where it was overall pretty smooth, but it would take awhile to get to the desired distance.

d. (5 points) List the third set of gain K_P and K_D, and clamping ranges, and describe the robot's behavior.

Next we did Kp 20 and Kd 10 with clamping ranges of -10 and 10. Overall this one did the best staying at the desired distance on the turns but had by far the most oscillations and was the most gittery. When upping the cmaping to -20 and 20 and was basically the same but even more jittery.

e. (5 points) Best gain K_P and K_D Kp 5 and Kd 1

f. (5 points) P vs. PD Controller

The PD Controller was less jittery overall compared to the P controller and was also better during the turns.

3.3 Faster Wall Following

a. (5 points) Describe the behavior of your robot when using the previous gain values with your new speed.

The robot overcompansated on the turns, going wide then in too much getting very close to the wall after finishing the turn so it would then need to go back out. We figured out afterward this was because our clamping values of -10 and 10 were too tight so when it went on a turn it did not turn hard enough.

b. (10 points) Wall following at 200 base speed [Video]

c. (5 points) Best gain K_P and K_D Kp 6 snd Kd 3, with -15 and 15 clamping